

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1290

Gwinn 138 kV
May 15, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1290. The Interconnection Customer proposes a 144.0 MW solar plus storage facility interconnecting at the Gwinn 138 kV substation in the Pennsylvania area. The request was received on September 13, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 379 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$1,172,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Nodaway-Sunrise Ctr Substation 138 kV is overloaded to 100.6% of its 281 MVA emergency rating for the loss of the Lutesville-Edes Tap 2 345 kV Line. The project contributes 22.2% DFAX (31.9 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sunrise Ctr-Nodaway 138 kV	Reconductor line	\$1,948,000	60% (\$1,172,000)
	Total allocated to project		\$1,172,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.